Junghan Kim

Applied AI / Agent Platform Engineer — Accountable Systems Around Models

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Summary

I build and operate the systems around models: harness boundaries, retrieval, permissions, evidence contracts, upgrade policy, and the work surface through which a human takes responsibility. I have operated a self-hosted multi-agent harness across repeated upstream changes, separated semantic memory into its own hybrid-retrieval layer, and published a dispatch substrate that preserves each harness's transcript, authentication, and runtime ownership.

My claim is not foundation-model training or fine-tuning expertise. It is applied AI systems engineering grounded in layers that must work when the model call ends: Go and Clojure services, PostgreSQL/pgvector, NixOS and containers, embedded Linux, protocols, and real domain data. I design where an agent may act, what it must leave behind, where it must stop, and how another human or agent can reproduce the path.

Core Competencies

- **Agent Platform Operations** — upstream tracking, staged promotion, regression instrumentation, rollback policy, SLOs, and recoverable session continuity.
- **Harness and Agent Systems** — dispatch, identity, auth boundaries, delegation contracts, domain-owner agents, and review loops with a human as PM.
- **Knowledge Engineering / RAG** — embedding and hybrid retrieval, reranking and embedding served in-house, cross-lingual expansion, recall tracking, Org/Denote knowledge graphs.
- **Backend and Infrastructure** — Go and Clojure services, PostgreSQL/pgvector, Docker, declarative NixOS, GPU infrastructure, reproducible build and document pipelines.
- **Embedded and Edge AI** — on-device sLLM and NPU deployment on ARM/RISC-V Linux; the agent loop is designed by someone who has had to make the system beneath it work.

Public Evidence — Inspect, Run, Challenge

- **Harness boundary** — github.com/junghan0611/entwurf publishes the package, boundary contract, deterministic checks, and clean-host path. Run `npm install -g @junghanacs/entwurf`, then `entwurf check-bridge`. An outside contributor exercised the extension boundary in PR #40.
- **Independent memory layer** — github.com/junghan0611/andenken exposes indexing, hybrid search, quality regression, and gap diagnosis through `./run.sh search:md`, `golden`, and `doctor --md`.
- **Reproducible skill surface** — github.com/junghan0611/agent-config fans one skill SSOT into six harness surfaces. Built CLI artifacts record source revision, source-tree digest, and binary SHA-256; `./run.sh env` reports drift. Private company data is not presented as public proof.

Experience

GoQual Inc. — Full Stack Architect 2025.06 – Present

Domain-Owner Agents and Agent Work Surfaces (2026.05 – Present)

The request I keep receiving from other teams is not "build me a dashboard." It is "make this run without depending on me." When a dashboard would merely transfer the burden, I stand up an agent and attach it to the person who owns the domain.

- **VOC workbench** — quantitative-abduction tool over daily customer-support data. Fixes the units, periods, and inclusion policy of every number; makes each figure traceable down to a single conversation id; surfaces signals that differ from baseline. The production counterpart of the public `abductcli` method, without exposing company data.
- **Incident workbench** — read-only evidence tool that reads live from real sources instead of storing dumps, aligning platform DDL catalog, device logs, and runtime mirror on a single KST time axis. Stops rather than warns when the evidence contract breaks.
- **Forge connector** — agents turning conversations into durable, reviewable work items on self-hosted Forgejo: issue, label, first review, hand-back. The bot is a dispatcher and recorder, never the implementer. Public as `forge-config`.
- **Chief-of-staff agent** — working surface carrying tracks, owners, decisions, and a ledger. **Product security response (PSIRT)** — intake, reproduction, impact assessment, vendor escalation, and regulator/customer response for externally reported vulnerabilities.

AI Infrastructure and Knowledge Base (2025.06 – 2025.09)

- Built GPU cluster infrastructure: 3× RTX 5080, NixOS, 10G network, 17 Docker services.
- Designed a hierarchical AI agent system on n8n (40+ node workflows) over Supabase pgvector; served the embedding and reranking stages directly rather than renting them.
- Data pipelines: Airbyte connectors for Channel.io, Notion, and JIRA into PostgreSQL JSONB; internal portal behind Cloudflare Zero Trust.

On-Device AI Smart Home Hub (2025.09 – Present)

- **HomeAgent** — open-source Matter hub with an on-device AI agent. One Go codebase runs on RPi5 + Hailo-8 NPU (Yocto) and RK3576 (Android), with Flutter as the shell. BLE commissioning for Matter/Thread via a matterjs bridge, OTBR integration. Ran three agents in parallel with the human as PM; the work split held — no file conflicts across the sessions.
- **Matter wallpad (customer engagement)** — RK3576 + Android 15, ESP32-H2 as Thread RCP, AOSP-native CHIP C++ SDK with `ot-daemon`, delivered as a versioned Android SDK (AAR) into the customer's own namespace.
- **National R&D (IITP, 2025 – 2028)** — intelligent-home program with an AI SoC vendor. Our scope: sLLM-driven voice control, Matter integration, and porting the NPU workload across accelerator families. Currently in year two.

Smart Home Hub — Firmware to Product (2025.12 – Present)

- **SKS Hub** — Zigbee/Wi-Fi gateway firmware in Zig, **shipped to mass production**. Rewrote a legacy C Zigbee SDK integration behind Zig's type-safe FFI layer; deterministic state machine (single `HubState`, pure `transition()`) across driver, protocol, and cloud layers; 24-hour aging-test automation with an AI agent as pair programmer.
- **Productization the original contract did not cover** — built the server (Go) and companion app (Flutter) myself, turning a firmware deliverable into a whole product. One protocol served by two interchangeable backends (AWS IoT for connected deployments, a local mTLS broker for closed networks) with the firmware shipping unmodified in both; multi-hub fan-out verified on real hardware; shadow mirroring with monotonic versioning.
- **Next-generation hub** — moved the port target from ARMv7/glibc with a vendor sysroot to RISC-V (SG2000) on statically linked musl; extracted a board HAL so the hub core no longer knows what board it runs on; delegated Zigbee transport to Zigbee2MQTT (EFR32MG24); went device-agnostic by dropping seven hardcoded device-type handlers.

Independent Work — Operating an Agent Platform in the Open — 2022 – Present

Public work is not a side project here — it is where the ideas are built and load-tested before they enter production settings. One vertical runs through it: **operate a harness → extract the memory layer → make harnesses addressable → close the development loop.**

Agent Platform Operations (2026.02 – Present)

Operated a self-hosted multi-agent harness across repeated upstream release changes on Oracle Cloud ARM. Instrumented a latency regression to a spinning gateway process, chose the rollback target on evidence rather than recency, and wrote the policy that followed: no two-version jumps, stage on a non-production agent for ≥24h before promotion, responsiveness is the SLO. A later upstream release fixed the root cause, and the policy is what decided when to take it. The incident, the rollback target, and the policy that followed are dated in the public commit history.

entwurf — Garden-Citizen Dispatch Substrate

Agent harnesses (Claude Code, Codex, Antigravity, pi) address one another by id **without** owning each other's transcript, auth, or runtime. No OAuth proxy, no CLI transcript scraping, no backend identity replacement. Published on npm.

andenken — Semantic Memory as an Independent Layer

LanceDB with hybrid retrieval (vector + FTS, score-normalized), Korean↔English cross-lingual expansion through a personal vocabulary graph, and recall tracking for consolidation.

agent-config — Reproducible Skill Surface

PKM-native stack of 40+ agent skills over notes, agenda, bibliography, health data, and git history, plus a 3-layer cross-lingual retrieval design. Built CLI artifacts carry source revision, source-tree digest, and installed-binary SHA-256 provenance.

forge-config — Durable Agent Work Ledger

Public operating policy and CLI surface that turns conversations into reviewable Forgejo issues: capture, explicit ownership, read-only first review, hand-back, and human-governed implementation. The bot dispatches and records; it does not silently implement.

Agent-Queryable Corpus Tools

Built `denotecli`, `dictcli`, `gitcli`, `lifetract`, `bibcli`, and `abductcli` for a real corpus rather than a demo context. Every tool speaks Denote IDs, so commits, journals, bot logs, and health days cross-reference on one timestamp scheme.

Sungkyunkwan University — Distributed Computing Lab 2018 – 2022

- Research on non-volatile memory filesystems and NUMA lock performance in virtualized environments.
- Virginia Tech COSMOSS Lab, exchange researcher (2019.07 – 2020.03).

NEMO-UX — Co-founder 2013 – 2017

- Built a Linux-based large-format touch display OS for commercial and education markets.
- Full embedded product lifecycle: hardware integration, OS customization, application development, and mass production. The startup failed; the habit of taking a product from silicon to shipping did not.

Third-Party Validation

Evidence that does not depend on my own account — each item is an action someone else took.

- **Others build on entwurf.** A developer I have never met contributed a Snowflake Cortex Code ACP backend (entwurf#40 — 11 files, +885 lines), targeting an enterprise agent runtime I had never written for. He found the extension boundary exactly where the architecture claimed it was — stronger validation than a self-authored architecture claim.
- **Others merged my code.** Two patches accepted upstream into dakra/ghostel, an Emacs terminal emulator built on libghostty-vt: #343 forwards IME-committed text and guards redraws during composition (+435 lines), and #510 lets Lisp IMEs compose inside read-only buffers. Both fix Korean and CJK input in a codebase I do not own — the same class of problem `memex-kb` exists to solve, met from the other side.
- **People install it.** `@junghanacs/entwurf` — 1,395 npm installs in the preceding 30 days (2026-07).

Technical Stack

Domain	Technologies
Languages	Go · Clojure · Zig · C · TypeScript · Elisp · Nix · Bash · Python
Agent and AI	Agent harnesses (Claude Code, Codex, pi) · ACP · MCP · A2A · embedding retrieval · LanceDB · hybrid RAG · sLLM workflows · NPU deployment · Ollama
Embedded and IoT	ARM Linux · RISC-V · Yocto · static musl · board HAL · Matter · Thread · Zigbee · Zigbee2MQTT · MQTT · OTBR · AWS IoT · mTLS
Backend and Infra	NixOS · Docker · GPU cluster · PostgreSQL · pgvector · n8n · Airbyte · Cloudflare Zero Trust · GraalVM native-image
Interface and Docs	Flutter · React · A2UI · SSE · JSON-RPC 2.0 · REST · Emacs/Org-mode · Denote · Pandoc · `XeLaTeX`

Education and Awards

- **Sungkyunkwan University** — Ph.D. coursework completed (2010 – 2012) and M.S. (2008 – 2010), Computer Science · **Sejong University** — B.S., Computer Science (2004 – 2008)
- **Prime Minister's Award**, Korea Software Competition (2010) — mobile virtualization software